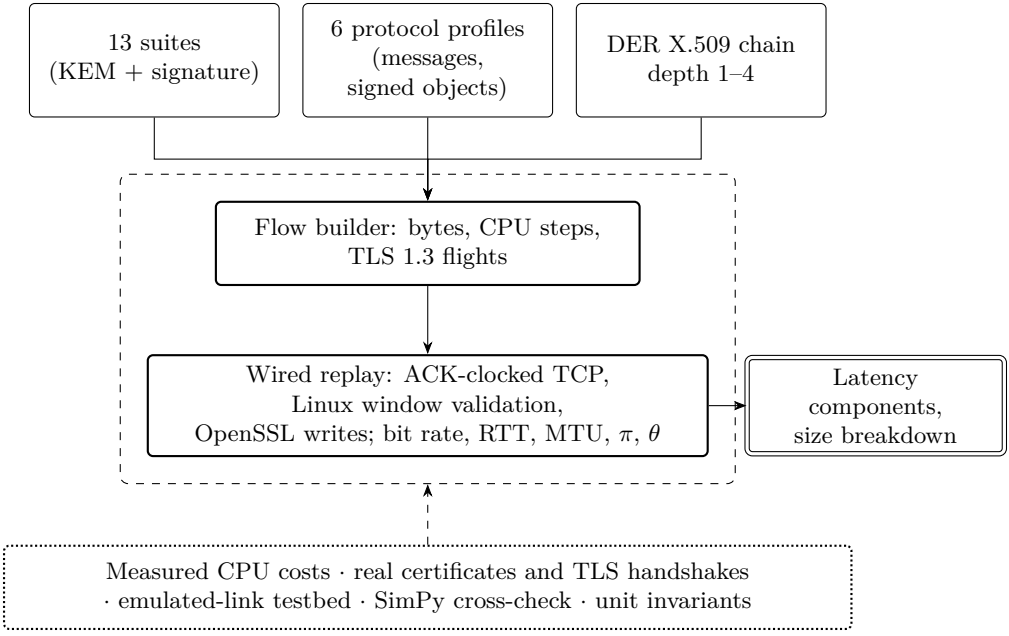
13 suites
(KEM + signature)
6 protocol profiles
(messages,
signed objects)
DER X.509 chain
depth 1–4
Flow builder: bytes, CPU steps,
TLS 1.3 flights
Wired replay: ACK-clocked TCP,
Linux window validation,
OpenSSL writes; bit rate, RTT, MTU, π, θ
Latency
components,
size breakdown
Measured CPU costs · real certificates and TLS handshakes
· emulated-link testbed · SimPy cross-check · unit invariants